MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Antimicrobial Dosing Recommendations for Pediatric Patients (≥ 2 months of age and post-menstrual age > 44 weeks)

These guidelines include the most common indications for antimicrobial use at Michigan Medicine. Dosing recommendations are based on estimated GFR using the modified Schwartz equation[1], which has not been validated in children < 1 year of age. Contact your team's pharmacist (or the pharmacy at 764-8208) to discuss appropriate dosage adjustments in children < 1 year of age with impaired renal function. Please use Lexicomp® for dosing in other indications. Refer to the Antimicrobial Stewardship Team website for additional treatment guidelines of infections in pediatric patients.

Antibiotics			
Amikacin IV	Ceftaroline IV	Daptomycin IV	Nafcillin IV
Amoxicillin PO	Ceftazidime IV	Doxycycline IV/PO	Nitrofurantoin PO
Amoxicillin-clavulanate PO	Ceftazidime-avibactam IV	Ertapenem IV	Penicillin G IV
Ampicillin IV	Ceftolozane-tazobactam IV	Gentamicin IV	Piperacillin-tazobactam IV
Ampicillin-sulbactam IV	Cefiderocol IV	Imipenem IV	Rifampin IV/PO
Azithromycin IV/PO	Ceftriaxone IV	Imipenem-relebactam IV	Tigecycline IV
Aztreonam IV	Cefuroxime axetil PO	Levofloxacin IV/PO	TMP-SMX IV/PO
Cefazolin IV	Cefuroxime IV	Linezolid IV/PO	Tobramycin IV
Cefepime IV	Cephalexin PO	Meropenem IV	Vancomycin IV
Cefixime PO	Ciprofloxacin PO	Meropenem-vaborbactam IV	
Cefoxitin IV	Ciprofloxacin IV	Minocycline IV/PO	
Cefpodoxime PO	Clindamycin IV/PO	Metronidazole IV/PO	
Antifungals			
Amphotericin B, liposomal IV	Itraconazole PO	Pentamidine IV	Voriconazole IV/PO
Fluconazole IV/PO	Micafungin IV	Posaconazole IV/PO	
Antivirals			
Acyclovir IV	Foscarnet IV	Peramivir IV	
Baloxavir PO	Ganciclovir IV	Valganciclovir PO	
Cidofovir IV	Oseltamivir PO	Zanamivir INH	
Notes			
Footnotes		References	

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
ACYCLOVIR IV							
Mucocutaneous HSV infection (mild AND immunocompetent, all ages)	5 mg/kg/dose q8h	5 mg/kg/dose q12h	5 mg/kg/dose q24h	2.5 mg/kg/dose q24h	5 mg/kg/dose q24h		5 mg/kg/dose q8h
Mucocutaneous HSV infection (moderate to severe OR immunocompromised, all ages)	10 mg/kg/dose q8h	10 mg/kg/dose q12h	10 mg/kg/dose q24h	5 mg/kg/dose q24h			10 mg/kg/dose q8h
0 to 3 months old	20 mg/kg/dose q8h	20 mg/kg/dose q12h	20 mg/kg/dose q24h	10 mg/kg/dose q24h			20 mg/kg/dose q8h
Disseminated HSV or HSV encephalitis (3 months to < 12 years)	15 mg/kg/dose q8h	15 mg/kg/dose q12h	15 mg/kg/dose q24h	7.5 mg/kg/dose q24h			15 mg/kg/dose q8h
Disseminated HSV or HSV encephalitis (≥ 12 years)	10 mg/kg/dose q8h	10 mg/kg/dose q12h	10 mg/kg/dose q24h	5 mg/kg/dose q24h			10 mg/kg/dose q8h
Varicella infections (chickenpox, shingles)							
AMIKACIN[4] IV							
	See Empiric Dosing and Monitoring Recommendations for Aminoglycosides in Pediatric Patients						
AMOXICILLIN PO							
Otitis media/PCN-resistant *S. pneumoniae*	45 mg/kg/dose q12h (max: 2 g/dose)		20 mg/kg/dose q12h	20 mg/kg/dose q24h (max: 1 g/dose)			No information
Community-acquired pneumonia	45 mg/kg/dose q12h (max: 2 g/dose)		20 mg/kg/dose q12h				
Other infections	25 mg/kg/dose q12h (max: 2 g/dose)		20 mg/kg/dose q12h				

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
AMOXICILLIN-CLAVULANATE PO	Dosing based on amoxicillin component (Click here for formulations based on amoxicillin:clavulanate ratio)						
Otitis media, rhinosinusitis, outpatient pneumonia	45 mg/kg/dose q12h (max: 2 g/dose) (amox:clav ratio 14:1)		20 mg/kg/dose q12h (max: 1 g/dose) (amox:clav ratio 7:1)	20 mg/kg/dose q24h (max: 500 mg/dose) (amox:clav ratio 4:1)			45 mg/kg/dose q12h (max: 2 g/dose) (amox:clav ratio 14:1)
Inpatient pneumonia	30 mg/kg/dose q8h (max: 1 g/dose) (amox:clav ratio 14:1)		20 mg/kg/dose q12h (max: 1 g/dose) (amox:clav ratio 7:1)	20 mg/kg/dose q24h (max: 500 mg/dose) (amox:clav ratio 4:1)			30 mg/kg/dose q8h (max: 1 g/dose) (amox:clav ratio 14:1)
Bone and joint infections	33 mg/kg/dose q8h (max: 1 g/dose) (amox:clav ratio 14:1)		20 mg/kg/dose q12h (max: 1 g/dose) (amox:clav ratio 7:1)	20 mg/kg/dose q24h (max: 500 mg/dose) (amox:clav ratio 4:1)			33 mg/kg/dose q8h (max: 1 g/dose) (amox:clav ratio 14:1)
Intra-abdominal infections	15 mg/kg/dose q8h (max: 500 mg/dose) (amox:clav ratio 4:1)		20 mg/kg/dose q12h (max: 500 mg/dose) (amox:clav ratio 4:1)	20 mg/kg/dose q24h (max: 500 mg/dose) (amox:clav ratio 4:1)			15 mg/kg/dose q8h (max: 500 mg/dose) (amox:clav ratio 4:1)
Urinary tract infections	25 mg/kg/dose q12h (max: 875 mg/dose) (amox:clav ratio 7:1)		20 mg/kg/dose q12h (max: 500 mg/dose) (amox:clav ratio 7:1)	20 mg/kg/dose q24h (max: 500 mg/dose) (amox:clav ratio 4:1)			25 mg/kg/dose q12h (max: 875 mg/dose) (amox:clav ratio 7:1)
AMPHOTERICIN B, LIPOSOMAL IV							
Invasive fungal infections	5 mg/kg/dose q24h						
AMPICILLIN IV							
Community-acquired pneumonia:							
H influenzae (beta-lactamase negative), Group A *Streptococcus*	50 mg/kg/dose q6h (max: 2 g/dose)		50 mg/kg/dose q8h	50 mg/kg/dose q12h			50 mg/kg/dose q6h
S pneumoniae (MIC ≤ 2 mcg/mL)	50 mg/kg/dose q6h (max: 2 g/dose)		50 mg/kg/dose q8h	50 mg/kg/dose q12h			50 mg/kg/dose q6h
S pneumoniae (MIC ≥ 4 mcg/mL)	100 mg/kg/dose q6h (max: 2 g/dose)		100 mg/kg/dose q8h	100 mg/kg/dose q12h	50 mg/kg/dose q12h	100 mg/kg/dose q12h	100 mg/kg/dose q6h
Urinary tract infections	50 mg/kg/dose q6h (max: 2 g/dose)		50 mg/kg/dose q8h	50 mg/kg/dose q12h			50 mg/kg/dose q6h
Endocarditis, meningitis	100 mg/kg/dose q6h (max: 3 g/dose)		100 mg/kg/dose q8h	100 mg/kg/dose q12h	50 mg/kg/dose q12h	100 mg/kg/dose q12h	100 mg/kg/dose q6h
AMPICILLIN-SULBACTAM IV	Dosing based on ampicillin component – For maximum doses, 2 g ampicillin/dose = 3 g ampicillin-sulbactam/dose OR 3 g ampicillin/dose = 4.5 g ampicillin-sulbactam/dose						
Most indications, cystic fibrosis exacerbation	50 mg/kg/dose q6h (max: 2 g ampicillin/dose)		50 mg/kg/dose q8h	50 mg/kg/dose q12h			50 mg/kg/dose q6h
Endocarditis, meningitis, *Acinetobacter* infections	100 mg/kg/dose q6h (max: 3 g ampicillin/dose)		100 mg/kg/dose q8h	100 mg/kg/dose q12h			100 mg/kg/dose q6h
AZITHROMYCIN IV/PO							
Most indications	10 mg/kg/dose load (max: 500 mg/dose), then 5 mg/kg/dose q24h (max: 250 mg/dose)						
AZTREONAM IV							

		CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
CNS infections		50 mg/kg/dose q6h (max: 2 g/dose)		33 mg/kg/dose q8h (max: 2 g/dose)	20 mg/kg/dose q12h (max: 2 g/dose)			50 mg/kg/dose q6h (max: 2 g/dose)
Systemic infections, including sepsis, cystic fibrosis exacerbation, and febrile neutropenia		50 mg/kg/dose q8h (max: 2 g/dose)		20 mg/kg/dose q8h (max: 2 g/dose)	10 mg/kg/dose q12h (max: 1 g/dose)			50 mg/kg/dose q8h (max: 2 g/dose)
BALOXAVIR PO								
Treatment and prophylaxis (duration = 1 days)								
<3 months of age		1 mg/kg x 1 dose						
≥ 3 months old	< 20 kg	2 mg/kg x 1 dose						
	20- < 80 kg	40 mg x 1 dose						
	≥ 80 kg	80 mg x 1 dose						
CEFAZOLIN IV								
Cystitis		17 mg/kg/dose q8h (max: 1 g/dose)		17 mg/kg/dose q12h	17 mg/kg/dose q24h	25 mg/kg/dose q24h		17 mg/kg/dose q8h
Endocarditis, systemic infections, skin and soft tissue infections		33 mg/kg/dose q8h (max: 2 g/dose)		33 mg/kg/dose q12h	33 mg/kg/dose q24h			33 mg/kg/dose q8h
Osteomyelitis, MSSA pneumonia, cystic fibrosis exacerbation		50 mg/kg/dose q8h (max: 2 g/dose)		50 mg/kg/dose q12h	50 mg/kg/dose q24h			50 mg/kg/dose q8h
CEFEPIME IV		Extended infusion over 4 hours is preferred for most patients						
Systemic infections, including cystic fibrosis exacerbation		50 mg/kg/dose q8h (max: 2 g/dose)	50 mg/kg/dose q12h (max: 2 g/dose)	50 mg/kg/dose q24h (max: 2 g/dose)	50 mg/kg/dose q24h (max: 1 g/dose)	50 mg/kg/dose q24h (max: 2 g/dose)	25 mg/kg/dose q24h (max: 1 g/dose) OR 50 mg/kg/dose post HD 3x/week (max: 2 g/dose)	50 mg/kg/dose q8h (max: 2 g/dose)
CEFIDEROCOL IV		Extended infusion over 3 hours is preferred for most patients						
All indications (*Consult ID PharmD*)		60 mg/kg/dose q8h (max: 2 g/dose)	45 mg/kg/dose q8h (max: 1.5 g/dose)	30 mg/kg/dose q8h (max: 1 g/dose)	22.5 mg/kg/dose q12h (max: 750 mg/dose)			60 mg/kg/dose q8h (max: 2 g/dose)
CEFIXIME PO								
Most indications		4 mg/kg/dose q12h (max: 200 mg/dose)		4 mg/kg/dose q24h		No information, not significantly removed		
CEFOXITIN IV								
Urinary tract infections		20 mg/kg/dose q6h (max: 1 g/dose)	20 mg/kg/dose q8h	20 mg/kg/dose q12h	20 mg/kg/dose q24h			20 mg/kg/dose q6h
Intra-abdominal infections, pelvic inflammatory disease		40 mg/kg/dose q6h (max: 2 g/dose)	40 mg/kg/dose q8h	40 mg/kg/dose q12h	40 mg/kg/dose q24h			40 mg/kg/dose q6h

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
Mycobacterial infections, with or without cystic fibrosis exacerbation	50 mg/kg/dose q6h (max: 3 g/dose)	50 mg/kg/dose q8h	50 mg/kg/dose q12h	50 mg/kg/dose q24h			50 mg/kg/dose q6h
CEFPODOXIME PO							
Most indications	5 mg/kg/dose q12h		5 mg/kg/dose q24h			5 mg/kg/dose 3x/week	No information
CEFTAROLINE IV							
2-5 months: Systemic infections, including cystic fibrosis exacerbation	10 mg/kg/dose IV q8h	8 mg/kg/dose IV q8h	6 mg/kg/dose IV q8h	4 mg/kg/dose IV q8h			10 mg/kg/dose IV q8h
≥ 6 months: Systemic infections, including cystic fibrosis exacerbation	15 mg/kg/dose IV q8h (max: 600 mg/dose)	10 mg/kg/dose IV q8h (max: 400 mg/dose)	8 mg/kg/dose IV q8h (max: 300 mg/dose)	5 mg/kg/dose IV q8h (max: 200 mg/dose)			15 mg/kg/dose IV q8h (max: 600 mg/dose)
CEFTAZIDIME IV	Extended infusion over 4 hours is preferred for most patients						
Systemic infections, CNS infections, cystic fibrosis exacerbation	50 mg/kg/dose q8h (max: 2 g/dose)	50 mg/kg/dose q12h	50 mg/kg/dose q24h	50 mg/kg/dose q48h			50 mg/kg/dose q8h
CEFTAZIDIME-AVIBACTAM IV	Dosing based on ceftazidime component – For maximum doses, 2 g ceftazidime/dose = 2.5 g ceftazidime-avibactam/dose Extended Infusion over 3 hours is preferred for most patients						
Systemic infections, CNS infections, cystic fibrosis exacerbation	50 mg/kg/dose q8h (max: 2 g ceftazidime/dose)	25 mg/kg/dose q8h (max: 1 g ceftazidime/dose)	19 mg/kg/dose q12h (max: 750 mg ceftazidime/dose)	19 mg/kg/dose q24h (max: 750 mg ceftazidime/dose)			50 mg/kg/dose q8h (max: 2 g ceftazidime/dose)
CEFTOLOZANE-TAZOBACTAM IV	Dosing based on ceftolozane component – For maximum doses, 2 g ceftolozane/dose = 3 g ceftolozane-tazobactam/dose Extended infusion over 3 hours is preferred for most patients						
CNS infections, pneumonia, cystic fibrosis exacerbation	40 mg/kg/dose q8h (max: 2g ceftolozane/dose)	20 mg/kg/dose q8h (max: 1g ceftolozane/dose)	10 mg/kg/dose q8h (max: 500 mg ceftolozane/dose)	30 mg/kg/dose (max: 1,500 mg of ceftolozane) x 1, then 6 mg/kg/dose q8h (max: 300 mg ceftolozane/dose)			40 mg/kg/dose q8h (max: 1g ceftolozane/dose
CEFTRIAXONE IV							
Urinary tract infections, non-severe community-acquired pneumonia	50 mg/kg/dose q24h (max: 2 g/dose)						
Intra-abdominal infections	75 mg/kg/dose q24h (max: 2 g/dose)						
Endocarditis	100 mg/kg/dose q24h (max: 2 g/dose)						
CNS infections, severe or complicated pneumonia, sepsis	50 mg/kg/dose q12h (max: 2 g/dose)						
CEFUROXIME AXETIL PO							
Most indications	15 mg/kg/dose q12h (max: 500 mg/dose)			15 mg/kg/dose q24h			15 mg/kg/dose q12h
CEFUROXIME IV							

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
Most indications	50 mg/kg/dose q8h (max: 2 g/dose)		50 mg/kg/dose q12h	50 mg/kg/dose q24h			50 mg/kg/dose q8h
CEPHALEXIN PO							
Urinary tract infections, skin and soft tissue infections	25 mg/kg/dose q8h (max: 1 g/dose)	10 mg/kg/dose q8h	10 mg/kg/dose q12h	10 mg/kg/dose q24h	10 mg/kg/dose q24h	10 mg/kg/dose q24h	25 mg/kg/dose q8h
MSSA community-acquired pneumonia	25 mg/kg/dose q6h (max: 1 g/dose)	25 mg/kg/dose q8h	25 mg/kg/dose q12h	25 mg/kg/dose q24h	20 mg/kg/dose q24h	25 mg/kg/dose q24h	25 mg/kg/dose q6h
Osteomyelitis	37.5 mg/kg/dose q6h (max: 1 g/dose)	37.5 mg/kg/dose q8h	37.5 mg/kg/dose q12h	37.5 mg/kg/dose q24h	20 mg/kg/dose q24h	37.5 mg/kg/dose q24h	37.5 mg/kg/dose q6h
CIDOFOVIR IV							
BK virus nephropathy (WITHOUT probenecid)	0.25 mg/kg/dose x1, then discuss repeat dosing with transplant ID team						
Adenovirus viremia (WITH probenecid & hydration)	5 mg/kg/dose x1, then discuss repeat dosing with transplant ID team						
CIPROFLOXACIN PO							
Urinary tract infections	10 mg/kg/dose q12h (max: 500 mg/dose)		10 mg/kg/dose q24h				10 mg/kg/dose q12h
Systemic infections, intra-abdominal infections, invasive *Pseudomonas* infections	15 mg/kg/dose q12h (max: 750 mg/dose)		15 mg/kg/dose q24h				15 mg/kg/dose q12h
Cystic fibrosis exacerbation	20 mg/kg/dose q12h (max: 1 g/dose)		20 mg/kg/dose q24h			15 mg/kg/dose q24h	20 mg/kg/dose q12h
CIPROFLOXACIN IV							
Urinary tract infections, intra-abdominal infections without risk of MDRO, systemic infections	10 mg/kg/dose q12h (max: 400 mg/dose)		10 mg/kg/dose q18h	10 mg/kg/dose q24h			10 mg/kg/dose q12h
Cystic fibrosis exacerbation, invasive *Pseudomonas* infections, empiric therapy for risk of MDRO infections, endocarditis	10 mg/kg/dose q8h (max: 400 mg/dose)		15 mg/kg/dose q18h	15 mg/kg/dose q24h			10 mg/kg/dose q8h
CLINDAMYCIN PO/IV							
Systemic infections, skin and soft tissue infections	10 mg/kg/dose q8h (max: PO 450 mg/dose; max: IV 900 mg/dose)						
Bone and joint infections	13 mg/kg/dose q8h (max: PO 600 mg/dose; max: IV 900 mg/dose)						
Group A *Streptococcus* pharyngitis	7 mg/kg/dose q8h (max: PO 300 mg/dose)						

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
DAPTOMYCIN IV							
Skin/soft tissue infections	1- < 2 years: 10 mg/kg/dose q24h 2-6 years: 9 mg/kg/dose q24h 7-11 years: 7 mg/kg/dose q24h 12-17 years: 5 mg/kg/dose q24h	1-< 2 years: 10 mg/kg/dose q24h 2-6 years: 9 mg/kg/dose q24h 7-11 years: 7 mg/kg/dose q24h 12-17 years: 5 mg/kg/dose q24h	1-< 2 years: 10 mg/kg/dose q48h 2-6 years: 9 mg/kg/dose q48h 7-11 years: 7 mg/kg/dose q48h 12-17 years: 5 mg/kg/dose q48h	1-< 2 years: 10 mg/kg/dose q48h 2-6 years: 9 mg/kg/dose q48h 7-11 years: 7 mg/kg/dose q48h 12-17 years: 5 mg/kg/dose q48h	1-< 2 years: 10 mg/kg/dose q48h 2-6 years: 9 mg/kg/dose q48h 7-11 years: 7 mg/kg/dose q48h 12-17 years: 5 mg/kg/dose q48h	1-< 2 years: 10 mg/kg/dose q48h 2-6 years: 9 mg/kg/dose q48h 7-11 years: 7 mg/kg/dose q48h 12-17 years: 5 mg/kg/dose q48h	1-< 2 years: 10 mg/kg/dose q24h 2-6 years: 9 mg/kg/dose q24h 7-11 years: 7 mg/kg/dose q24h 12-17 years: 5 mg/kg/dose q24h
MRSA/VRE bacteremia, endocarditis, osteomyelitis	10-12 mg/kg/dose q24h	10-12 mg/kg/dose q24h	10-12 mg/kg/dose q48h	10-12 mg/kg/dose q48h	10-12 mg/kg/dose q48h	10-12 mg/kg/dose q48h	10-12 mg/kg/dose q24h
DOXYCYCLINE IV/PO							
Most indications, including cystic fibrosis exacerbation	2.2 mg/kg/dose q12h (max: 100 mg/dose)						
ERTAPENEM IV							
Most indications	15 mg/kg/dose q12h (Infants/children max: 500 mg/dose; Adolescents max: 1 g/dose)		15 mg/kg/dose q24h				15 mg/kg/dose q12h
FLUCONAZOLE IV/PO							
Urinary tract infections	3 mg/kg/dose q24h (max: 200 mg/dose)	1.5 mg/kg/dose q24h		1.5 mg/kg/dose q48h			3 mg/kg/dose q24h
Oropharyngeal, esophageal candidiasis	6 mg/kg/dose q24h (max: 400 mg/dose)	3 mg/kg/dose q24h		3 mg/kg/dose q48h			6 mg/kg/dose q24h
Systemic infections, CNS infections	12 mg/kg/dose q24h (max: 800 mg/dose)	6 mg/kg/dose q24h		6 mg/kg/dose q48h			12 mg/kg/dose q24h
FOSCARNET IV	mL/kg/min = [(0.413 x Ht in cm) ÷ SCr] ÷ Wt in kg						
	> 1.4 mL/kg/min	**1.4-1 mL/kg/min**	**1-0.8 mL/kg/min**	**0.79-0.6 mL/kg/min**	**0.59-0.5 mL/kg/min**	**0.49-0.4 mL/kg/min**	**< 0.4 mL/kg/min**
Induction, CMV	60 mg/kg/dose q12h	45 mg/kg/dose q12h	30 mg/kg/dose q12h	50 mg/kg/dose q24h	35 mg/kg/dose q24h	30 mg/kg/dose q24h	Not recommended
Maintenance, CMV	90 mg/kg/dose q24h	70 mg/kg/dose q24h	50 mg/kg/dose q24h	80 mg/kg/dose q48h	60 mg/kg/dose q48h	50 mg/kg/dose q48h	Not recommended

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
GANCICLOVIR IV							
Congenital CMV	6 mg/kg/dose q12h	3 mg/kg/dose q24h	1.5 mg/kg/dose q24h	1.5 mg/kg/dose 3 times per week			6 mg/kg/dose q12h
Induction, CMV	5 mg/kg/dose q12h	2.5 mg/kg/dose q24h	2.5 mg/kg/dose q24h	1.25 mg/kg/dose 3 times per week			5 mg/kg/dose q12h
Maintenance, CMV	5 mg/kg/dose q24h	1.25 mg/kg/dose q24h	0.625 mg/kg/dose q24h	0.625 mg/kg/dose 3 times per week			5 mg/kg/dose q24h
GENTAMICIN[4] IV							
	See Empiric Dosing and Monitoring Recommendations for Aminoglycosides in Pediatric Patients						
IMIPENEM IV	Extended infusion over 1 hour is preferred for *Pseudomonas* infections						
Systemic infections other than those listed below	15 mg/kg/dose q6h (max: 500 mg/dose)	7.5 mg/kg/dose q8h (max 500 mg/dose)	7.5 mg/kg/dose q12h (max: 500 mg/dose)	7.5 mg/kg/dose q24h (max: 250 mg/dose)			7.5 mg/kg/dose q8h (max: 500 mg/dose)
Cystic fibrosis exacerbation, CNS infections	25 mg/kg/dose q6h (max: 1 g/dose)	12.5 mg/kg/dose q8h (max: 1 g/dose)	12.5 mg/kg/dose q12h (max: 1 g/dose)	12.5 mg/kg/dose q24h (max: 500 mg/dose)			12.5 mg/kg/dose q8h (max: 500 mg/dose)
IMIPENEM-RELEBACTAM IV	Dosing based on imipenem component – For maximum doses, 1 g imipenem/dose = 2.5 g imipenem-relebactam-cilastatin/dose Extended infusion over 1 hour is preferred for *Pseudomonas* infections						
Systemic infections other than those listed below	15 mg/kg/dose q6h (max: 500 mg imipenem/dose)	7.5 mg/kg/dose q8h (max 500 mg imipenem/dose)	7.5 mg/kg/dose q12h (max: 500 mg imipenem/dose)	7.5 mg/kg/dose q24h (max: 250 mg imipenem/dose)			7.5 mg/kg/dose q8h (max: 500 mg imipenem/dose)
Cystic fibrosis exacerbation, CNS infections	25 mg/kg/dose q6h (max: 1 g imipenem/dose)	12.5 mg/kg/dose q8h (max: 1 g imipenem/dose)	12.5 mg/kg/dose q12h (max: 1 g imipenem/dose)	12.5 mg/kg/dose q24h (max: 500 mg imipenem/dose)			12.5 mg/kg/dose q8h (max: 500 imipenem mg/dose)
ITRACONAZOLE PO							
Histoplasmosis, blastomycosis	5 mg/kg/dose q12h (max: 400 mg/dose)						
LEVOFLOXACIN IV/PO							
Urinary tract infections	< 5 years: 8 mg/kg/dose q12h (max: 375 mg/dose) ≥ 5 years: 8 mg/kg/dose q24h (max: 750 mg/dose)	< 5 years: 8 mg/kg/dose q12h (max: 375 mg/dose) ≥ 5 years: 8 mg/kg/dose q24h (max: 750 mg/dose)	8 mg/kg/dose q24h	8 mg/kg/dose q48h			< 5 years: 8 mg/kg/dose q12h (max: 375 mg/dose) ≥ 5 years: 8 mg/kg/dose q24h (max: 750 mg/dose)
Systemic infections, cystic fibrosis exacerbation	< 5 years: 10 mg/kg/dose q12h (max: 375 mg/dose) ≥ 5 years: 10 mg/kg/dose q24h (max: 750 mg/dose)	< 5 years: 10 mg/kg/dose q12h (max: 375 mg/dose) ≥ 5 years: 10 mg/kg/dose q24h (max: 750 mg/dose)	10 mg/kg/dose q24h	10 mg/kg/dose q48h			< 5 years: 10 mg/kg/dose q12h (max: 375 mg/dose) ≥ 5 years: 10 mg/kg/dose q24h (max: 750 mg/dose)

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
LINEZOLID IV/PO							
< 12 years:	10 mg/kg/dose q8h (max: 600 mg/dose)				10 mg/kg/dose q12h		10 mg/kg/dose q8h
≥ 12 years:	10 mg/kg/dose q12h (max: 600 mg/dose)						10 mg/kg/dose q12h
MEROPENEM IV	Extended infusion over 3 hours is preferred for most patients						
Systemic infections other than those listed below	20 mg/kg/dose q8h (max: 1 g/dose)	20 mg/kg/dose q12h (max: 1 g/dose)	10 mg/kg/dose q12h (max: 500 mg/dose)	10 mg/kg/dose q24h (max: 500 mg/dose)			20 mg/kg/dose q8h (max: 1 g/dose)
Suspected or proven *Acinetobacter* or *Pseudomonas* infections, cystic fibrosis exacerbation, febrile neutropenia, CNS infections, septic shock, burn infections	40 mg/kg/dose q8h (max: 2 g/dose)	40 mg/kg/dose q12h (max: 2 g/dose)	20 mg/kg/dose q12h (max: 1 g/dose)	20 mg/kg/dose q24h (max: 1 g/dose)			40 mg/kg/dose q8h (max: 2 g/dose)
MEROPENEM-VABORBACTAM IV	Dosing based on meropenem component – For maximum doses, 2 g meropenem/dose = 4 g meropenem-vaborbactam/dose Extended Infusion over 3 hours is preferred for most patients						
Systemic infections	40 mg/kg/dose q8h (max: 2 g meropenem /dose)	40 mg/kg/dose q12h (max: 2 g meropenem /dose)	20 mg/kg/dose q12h (max: 2 g meropenem /dose)	20 mg/kg/dose q24h (max: 1 g meropenem /dose)			40 mg/kg/dose q8h (max: 2 meropenem g/dose)
METRONIDAZOLE PO							
Mild-moderate *C. difficile* infection	7.5 mg/kg/dose q6h (max: 500 mg/dose)			4 mg/kg/dose q6h			7.5 mg/kg/dose q6h
Systemic infections, CNS infections	10 mg/kg/dose q8h (max: 500 mg/dose)			4 mg/kg/dose q6h			10 mg/kg/dose q8h
METRONIDAZOLE IV							
Intra-abdominal infections	10 mg/kg/dose q8h (max: 500 mg/dose) OR 30 mg/kg/dose q24h (max: 1.5 g/dose)			4 mg/kg/dose q6h			10 mg/kg/dose q8h OR 30 mg/kg/dose q24h
Fulminant *C difficile* infection	7.5 mg/kg/dose q6h (max: 500 mg/dose)			4 mg/kg/dose q6h			7.5 mg/kg/dose q6h
Systemic infections, CNS infections	10 mg/kg/dose q8h (max: 500 mg/dose)			4 mg/kg/dose q6h			10 mg/kg/dose q8h
MICAFUNGIN IV							
Prophylaxis	3 mg/kg/dose q24h (max: 50 mg/dose)						
Treatment	5 mg/kg/dose q24h (max: 150 mg/dose)						
MINOCYCLINE IV/PO							
Systemic infections	4 mg/kg/dose (max: 200 mg/dose) x 1, then 2 mg/kg/dose (max: 100 mg/dose) q12h						

		CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
Acinetobacter infections		4 mg/kg/dose (max: 200 mg/dose) q12h						
NAFCILLIN IV								
Systemic infections, CNS infections		37.5 mg/kg/dose q6h (max: 2 g/dose)		37.5 mg/kg/dose q6h adjustment may be required for concomitant hepatic failure				
Endocarditis		33 mg/kg/dose q4h (max: 2 g/dose)		33 mg/kg/dose q4h adjustment may be required for concomitant hepatic failure				
NITROFURANTOIN PO		**(monohydrate/macro-crystals capsules, Macrobid®)**						
Cystitis		100 mg q12h		Not recommended				
NITROFURANTOIN PO		**(suspension, Macrodantin®)**						
Cystitis								
7-11.9 kg		12.5 mg q6h		Not recommended				
12-21.9 kg		25 mg q6h						
22-30.9 kg		37.5 mg q6h						
31-41.9 kg		50 mg q6h						
≥ 42 kg		75 mg q6h						
OSELTAMIVIR PO								
Treatment (duration = 5 days)								
≤ 8 months		3 mg/kg/dose q12h (max: 30 mg/dose)		3 mg/kg/dose q24h (max: 30 mg/dose)	3 mg/kg/dose q48h (max: 30 mg/dose)		7.5 mg x1, then 7.5 mg post each HD session	3 mg/kg/dose q12h (max: 30 mg/dose)
9 months to 11 months		3.5 mg/kg/dose q12h (max: 30 mg/dose)		3.5 mg/kg/dose q24h (max: 30 mg/dose)	3.5 mg/kg/dose q48h (max: 30 mg/dose)		7.5 mg x1, then 7.5 mg post each HD session	3.5 mg/kg/dose q12h (max: 30 mg/dose)
12 months to 17 years	≤ 15 kg	30 mg q12h		30 mg q24h	30 mg q48h		7.5 mg x1, then 7.5 mg post each HD session	30 mg q12h
	> 15-23 kg	45 mg q12h		45 mg q24h	45 mg q48h		10 mg x1, then 10 mg post each HD session	45 mg q12h
	> 23-40 kg	60 mg q12h		60 mg q24h	60 mg q48h		15 mg x1, then 15 mg post each HD session	60 mg q12h
	> 40 kg	75 mg q12h		75 mg q24h	75 mg q48h		30 mg x1, then 30 mg post each HD session	75 mg q12h
Prophylaxis (duration = 7 days)								
< 3 months		Not recommended						
3 months to 8 months		3 mg/kg/dose q24h (max: 30 mg/dose)		3 mg/kg/dose q48h (max: 30 mg/dose)	3 mg/kg/dose 2x/week (max: 30 mg/dose)		7.5 mg x1, then 7.5 mg post each HD session	3 mg/kg/dose q24h
9 months to 11 months		3.5 mg/kg/dose q24h (max: 30 mg/dose)		3.5 mg/kg/dose q48h (max: 30 mg/dose)	3.5 mg/kg/dose 2x/week (max: 30 mg/dose)		7.5 mg x1, then 7.5 mg post each HD session	3.5 mg/kg/dose q24h (max: 30 mg/dose)
12 months to 17 years	≤ 15 kg	30 mg q24h		30 mg q48h	30 mg 2x/week		7.5 mg x1, then 7.5 mg post each HD session	30 mg q24h
	> 15-23 kg	45 mg q24h		45 mg q48h	45 mg 2x/week		10 mg x1, then 10 mg post each HD session	45 mg q24h
	> 23-40 kg	60 mg q24h		60 mg q48h	60 mg 2x/week		15 mg x1, then 15 mg post each HD session	60 mg q24h
	> 40 kg	75 mg q24h		75 mg q48h	75 mg 2x/week		30 mg x1, then 30 mg	75 mg q24h

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
						post each HD session	
PENICILLIN G IV							
Endocarditis	50,000 units/kg/dose q4h (max: 4 million units/dose)		50,000 units/kg/dose q8h	25,000 units/kg/dose q8h			50,000 units/kg/dose q4h
Pneumococcal or meningococcal meningitis	75,000 units/kg/dose q6h (max: 6 million units/dose)		75,000 units/kg/dose q8h	37,500 units/kg/dose q8h			75,000 units/kg/dose q6h
Group B *Streptococcus* meningitis	125,000 units/kg/dose q6h (max: 6 million units/dose)		125,000 units/kg/dose q8h	62,500 units/kg/dose q8h			125,000 units/kg/dose q6h
Systemic infections	37,500 units/kg/dose q6h (max: 2 million units/dose)		37,500 units/kg/dose q8h	18,750 units/kg/dose q8h			37,500 units/kg/dose q6h
PENTAMIDINE IV							
Treatment, *Pneumocystis* pneumonia	4 mg/kg/dose q24h		4 mg/kg/dose q36h	4 mg/kg/dose q48h			4 mg/kg/dose q24h
PERAMIVIR IV							
Treatment (hospitalized / high-risk) (duration ≤ 5 days)							
3 months to 6 months	10 mg/kg/dose q24h	2.5 mg/kg/dose q24h	1.6 mg/kg/dose q24h	1.6 mg/kg/dose x1, then 0.25 mg/kg/dose q24h		1.6 mg/kg/dose x1 then 1.6 mg/kg/dose 2h post each HD session	10 mg/kg/dose q24h
> 6 months to 5 years	12 mg/kg/dose q24h (max: 600 mg/dose)	3 mg/kg/dose q24h (max: 150 mg/dose)	2 mg/kg/dose q24h (max: 100 mg/dose)	2 mg/kg/dose x1 (max: 100 mg/dose), then 0.3 mg/kg/dose q24h (max: 15 mg/dose)		2 mg/kg/dose x1 (max 100 mg/dose) then 2 mg/kg/dose 2h post each HD session (max 100 mg/dose)	12 mg/kg/dose q24h (max: 600 mg/dose)
6 years to 17 years	12 mg/kg/dose q24h (max: 600 mg/dose)	2.5 mg/kg/dose q24h (max: 150 mg/dose)	1.6 mg/kg/dose q24h (max: 100 mg/dose)	1.6 mg/kg/dose x1 (max: 100 mg/dose), then 0.25 mg/kg/dose q24h (max: 15 mg/dose)		1.6 mg/kg/dose x1 (max 100 mg/dose) then 1.6 mg/kg/dose 2h post each HD session (max 100 mg/dose)	12 mg/kg/dose q24h (max: 600 mg/dose)
Treatment (uncomplicated)							
6 months to 17 years	12 mg/kg/dose x1 (max: 600 mg/dose)	4 mg/kg/dose x1 (max: 200 mg/dose)	2 mg/kg/dose x1 (max: 100 mg/dose)			2 mg/kg/dose x1 (max: 100 mg/dose) - give 2h after HD session if patient is receiving dialysis that day	12 mg/kg/dose x 1 (max: 600 mg/dose)
PIPERACILLIN-TAZOBACTAM IV	Dosing based on piperacillin component – For maximum doses, 4 g piperacillin/dose = 4.5 g piperacillin-tazobactam/dose Extended infusion over 3 hours is preferred for most patients						
Systemic infections, empiric therapy for febrile neutropenia	75 mg/kg/dose q6h (max: 4 g piperacillin/dose)		75 mg/kg/dose q8h	75 mg/kg/dose q12h			75 mg/kg/dose q6h
Cystic fibrosis exacerbation	100 mg/kg/dose q6h (max: 4 g piperacillin/dose)		75 mg/kg/dose q8h		75 mg/kg/dose q12h		100 mg/kg/dose q6h
POSACONAZOLE IV/PO							

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
	Refer to Bone Marrow Transplant Infection Prophylaxis or Aspergillosis/Mucormycosis Guidelines						
RIFAMPIN IV/PO							
Endocarditis	7 mg/kg/dose q8h (max: 300 mg/dose)						
TIGECYCLINE IV							
Systemic infections other than those listed below	2 mg/kg/dose (max: 100 mg/dose) x 1, then 1.2 mg/kg/dose (max: 50 mg/dose) q12h						
Acinetobacter infections	4 mg/kg/dose (max: 200 mg/dose) x 1, then 3 mg/kg/dose (max: 100 mg/dose) q12h						
TMP-SMX (BACTRIM®) IV/PO	Dosing based on trimethoprim component						
Most infections	6 mg/kg/dose q12h (max: 320 mg/dose)		3 mg/kg/dose q12h	3 mg/kg/dose q24h	5 mg/kg/dose q24h (max: 80 mg/dose)	3 mg/kg/dose q24h	5-6 mg/kg/dose q12h
Pneumocystis pneumonia	5 mg/kg/dose IV 6h OR 10 mg/kg/dose PO q12h (max: 320 mg/dose)		5 mg/kg/dose IV/PO q12h	5 mg/kg/dose IV/PO q24h(max: 80 mg/dose)			5 mg/kg/dose IV q6h OR 10 mg/kg/dose PO q12h
TOBRAMYCIN[4] IV							
	See Empiric Dosing and Monitoring Recommendations for Aminoglycosides in Pediatric Patients						
VALGANCICLOVIR PO							
Congenital CMV	16 mg/kg/dose q12h	8 mg/kg/dose q24h	8 mg/kg/dose q48h	Give IV ganciclovir 2.5 mg/kg/dose 3 times per week	No information	Give IV ganciclovir 2.5 mg/kg/dose 3 times per week	Give IV ganciclovir 2.5 mg/kg/dose q24h
Prophylaxis/treatment of CMV	Refer to appropriate transplant guidelines						
VANCOMYCIN[4] IV							
	See Empiric Dosing and Monitoring Recommendations for Vancomycin						
VORICONAZOLE IV/PO							
< 16 years[5]	9 mg/kg/dose q12h						
≥ 16 years[5]	6 mg/kg/dose q12h x2 doses, then 4 mg/kg/dose q12h						
ZANAMIVIR INHALED							
Treatment (duration = 5 days)							
≥ 7 years	10 mg (2 inhalations) twice daily	10 mg (2 inhalations) twice daily	10 mg (2 inhalations) twice daily	10 mg (2 inhalations) twice daily	10 mg (2 inhalations) daily	10 mg (2 inhalations) daily	10 mg (2 inhalations) twice daily
< 7 years	Not recommended						
Prophylaxis (duration = 7 days)							

	CrCl[1] > 50 mL/min	CrCl[1] 50-30 mL/min	CrCl[1] 29-10 mL/min	CrCl[1] < 10 mL/min	Peritoneal Dialysis	Hemodialysis[2]	CRRT[3] Based on eGFR > 50 mL/min
≥ 5 years	10 mg (2 inhalations) daily	10 mg (2 inhalations) daily	10 mg (2 inhalations) daily	10 mg (2 inhalations) daily	10 mg (2 inhalations) daily	10 mg (2 inhalations) daily	10 mg (2 inhalations) daily
< 5 years	Not recommended						

Footnotes:

1. Modified Schwartz equation: $CrCl\ (ml/min) = \frac{0.413 * ht\ (cm)}{SCr\ (mg/dL)}$
2. Patients receiving Hemodialysis should have antimicrobials administered after dialysis on dialysis days.
3. CRRT = Continuous renal replacement therapy; For patients on CRRT: $eGFR\ (mL/min) = D_{rate}(mL/h) * \frac{1\ h}{60\ min} * \frac{1.73\ m^2}{BSA\ in\ m^2}$; dose antibiotic based on corresponding calculated CrCl
4. All patients on vancomycin or an aminoglycoside are evaluated by a pharmacist daily. For kinetics consultation with a pediatric pharmacist regarding special patient populations (e.g., those on ECMO or CRRT, or those with renal insufficiency), please contact the pharmacy at (76)4-8208.
5. Recommend oral formulation; IV vehicle (SCECD) can accumulate when CrCl < 50 mL/min. Only use IV formulation if benefit/risk to patient justifies use.

Amoxicillin: Clavulanate Formulations			
Amox:clav ratio	**Tablet**	**Suspension**	**Comments**
4:1	500-125 mg	250-62.5 mg/5 mL *or* 125-31.25 mg/5 mL	Clavulanate doses should not exceed 5 mg/kg/dose (125 mg/dose) or 10 mg/kg/day (375 mg/day) Tablets may be crushed, mixed with feeds, and/or administered through enteral tubes
7:1	875-125 mg	400-57 mg/5 mL *or* 200-28.5 mg/5 mL	
14:1	N/A	600-42.9 mg/5 mL	

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Revision History

Date	Revision
12/2020	Updated ceftazidime-avibactam dosing
03/2021	Updated aminoglycoside and vancomycin hyperlinks
02/2023	Updated aztreonam, cefepime, imipenem, meropenem, and piperacillin-tazobactam dosing
11/2023	Updated acyclovir, amoxicillin-clavulanate, doxycycline, oseltamivir, and peramivir dosing
09/2024	Added dosing for cefiderocol, ceftolozane-tazobactam, imipenem-relebactam, meropenem-vaborbactam, and tigecycline; updated dosing for ampicillin-sulbactam, cefoxitin, ceftaroline, ceftazidime-avibactam, and minocycline

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.